

PBL PROJECT – REVIEW 3

CASE-BASED PROBLEM STATEMENT

Project Title: Spam Email Classification Using Machine Learning

Project Domain: Machine Learning / Natural Language Processing

1. Case Scenario

In a modern organization, employees receive a large number of emails every day. These emails may include legitimate business communication as well as advertisements, promotional messages, fraudulent messages, phishing attempts, and other unwanted content. Manually checking every incoming email is time-consuming and may expose users to potentially harmful or deceptive messages.

To address this problem, the organization requires an automated email filtering system that can analyze the textual content of an incoming email and classify it as either **Spam** or **Ham (Legitimate)**. A machine learning-based solution can learn patterns from previously labelled emails and use those patterns to classify new, unseen emails.

2. Problem Statement

Develop a machine learning-based Spam Email Classification system that automatically classifies an email message as **Spam** or **Ham** based on its textual content. The system should preprocess the email data, perform exploratory data analysis, convert textual information into machine-readable features, train a suitable classification algorithm, and evaluate its performance using appropriate classification metrics.

3. Objectives

- To understand and analyze a labelled spam email dataset.
- To perform exploratory data analysis (EDA) and identify important patterns in the dataset.
- To preprocess email text by applying appropriate text-cleaning techniques.
- To convert textual data into numerical features using a suitable feature-extraction technique such as TF-IDF.
- To implement a machine learning classification algorithm such as Multinomial Naive Bayes.
- To evaluate the model using accuracy, precision, recall, F1-score, and a confusion matrix.
- To demonstrate the classification of new email messages as Spam or Ham.

4. Input and Output

Component	Description
Input	Text content of an email message.
Dataset	A labelled collection of email messages with Spam/Ham categories.
Processing	Text preprocessing, feature extraction, model training and classification.
Output	Predicted class: Spam or Ham .

5. Proposed Methodology

1. Collect and load the labelled spam email dataset.
2. Inspect the dataset for missing values, duplicates, class imbalance, and basic statistics.
3. Perform EDA using tables and visualizations.
4. Clean and preprocess the email text.
5. Split the dataset into training and testing sets.
6. Convert email text into numerical features using TF-IDF vectorization.
7. Train a classification model, such as Multinomial Naive Bayes.
8. Predict the classes of unseen test emails.
9. Evaluate the model using accuracy, precision, recall, F1-score, and confusion matrix.
10. Test the model with sample new email messages and display the predicted class.

6. Expected Outcome

The proposed system is expected to classify incoming email messages automatically and reduce the need for manual identification of unwanted emails. The project should demonstrate that machine learning and natural language processing techniques can be used to identify patterns in email text and provide reliable Spam/Ham predictions.

7. Review 3 Demonstration Requirements

- Case-based problem scenario and clearly defined problem statement.
- Practical implementation of dataset EDA using Python/Jupyter Notebook.
- Text preprocessing and feature extraction implementation.
- Algorithm demonstration and implementation.
- Model evaluation with appropriate performance metrics.
- Sample email prediction demonstrating Spam/Ham classification.

Note: This document is prepared as the case-based problem statement component for PBL Review 3.